BoMu: A Resistively-Loaded NMOS Differential Pair Design & Sizing (SKY130A, TinyTapeout)

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1 Introduction

`BoMu` is a single-stage differential amplifier built from a resistively-loaded NMOS source-coupled pair, designed for the SKY130A PDK and open-source TinyTapeout flow. Two matched NMOS devices (M_1, M_2) share a tail sink of $2I_D$, and each drain is pulled to the $1.2V$ supply through a load resistor R_D. The amplifier output is taken differentially across the drains, $v_{od} = v(\text{voutp}) - v(\text{voutn})$. A floating differential source $\pm v_{sd}/2$ on a common-mode V_{SC} drives the gates through 100Ω series resistors.

The driving bandwidth constraint is that each output drives an I/O pad of fixed $C_L = 5pF$. The tail sink is realised on-chip as a simple current mirror (Section 7.5) with a $200mV$ compliance budget. The flow is fully open-source (`xschem`, `ngspice`, `magic`/`klayout`); devices are sized by a g_m/I_D methodology [1] from measured current-density and speed data rather than an assumed square-law constant, and the small-signal results are standard [2].

1.1 Net identities

With the sign convention M_1 gate = `vinp` $\rightarrow$ drain = `voutn`, and M_2 gate = `vinn` $\rightarrow$ drain = `voutp`, the differential output $v_{od} = v(\text{voutp}) - v(\text{voutn})$ has positive gain with respect to the differential input $v_{id} = \text{vinp} - \text{vinn}$. The two half-circuits are identical by matching, so the analysis is done on a single common-source half-circuit and the differential result follows directly.

2 Specifications

The target specifications, and recommended design parameters, are collected in Table 1.

Table 1: Design targets and recommended parameters.

Parameter	**Value**	**Units**
Small-signal bandwidth f_{-3dB}	150	MHz
Differential DC gain A_{v0}	10	dB
Differential output swing v_{od}	400	mV_{pp}
Channel length L	0.15	μm
Supply V_{DD}	1.2	V (fixed)
Pad load C_L (each output)	5	pF (fixed)
Tail-sink compliance budget	200	mV (fixed)

3 First-Order Hand Design

The design proceeds as follows: the fixed pad capacitance and the bandwidth target set the load resistor, the gain target then sets the required transconductance, and a chosen inversion level (g_m/I_D) fixes the bias current.

Load resistor from bandwidth. The dominant pole sits at the output node, formed by the load resistance working into the fixed C_L. Ignoring the device output resistance for the moment,

$$R_D \;=\; \frac{1}{2\pi f_{-3dB} C_L} \;=\; \frac{1}{2\pi(150\text{MHz})(5\text{pF})} \;\approx\; 212\Omega. \tag{1}$$

Transconductance from gain. The differential (half-circuit) gain of a resistively-loaded pair is

$$A_{v0} \;=\; g_m(R_D \| r_o) \;\xrightarrow{r_o \gg R_D}\; g_m R_D. \tag{2}$$

With $A_{v0} = 10\text{dB} = 3.162\text{V/V}$ and Eq. (1),

$$g_m \;=\; \frac{A_{v0}}{R_D} \;=\; \frac{3.162}{212\Omega} \;\approx\; 14.9\text{mS}. \tag{3}$$

Bias current from g_m/I_D. Choosing a moderate-inversion $g_m/I_D = 15\text{V}^{-1}$ (a balance of device area, r_o, and speed at this minimum length) fixes the per-branch current and the tail current,

$$I_D \;=\; \frac{g_m}{(g_m/I_D)} \;=\; \frac{14.9\,\text{mS}}{15} \;\approx\; 0.99\text{mA}, \qquad I_{\text{tail}} = 2I_D \approx 2\text{mA}. \tag{4}$$

Swing and headroom. A $400mV_{pp}$ differential swing corresponds to a single-ended peak of $v_{od}/4 = 100mV$ about each drain's DC level $V_D = V_{DD} - I_D R_D \approx 0.99\,V$. Two conditions must hold: the bias drop must be large enough to source the swing, $I_D R_D > 100mV$ ($R_D = 212mV$ is ample), and the device must remain saturated at the bottom of the swing, $V_{D,\min} - (V_S + V_{ov}) > 0$, where V_S is the tail-node voltage held at or above the $200\,mV$ compliance budget. Both are comfortably satisfied because R_D is small and the supply headroom is large.

4 g_m/I_D Sizing Flow

Equations (1)–(3) fix R_D and the required g_m. The g_m/I_D method then turns the chosen inversion level into a physical width. A unit device is characterised over V_{GS} at fixed (V_{SB}, V_{DS}) to build a lookup grid returning, for any g_m/I_D, the current density J_D ($=I_D/W$), the intrinsic gain $g_m r_o$, and the transit frequency f_T. The width then follows as

$$W \;=\; \frac{I_D}{J_D(g_m/I_D,\, V_{SB},\, V_{DS})}. \tag{5}$$

This is implemented in `sim/sizing.py (worksheet)`, which references the `extern/sky130A-gmid` lookup grid (a git submodule). The script takes a `DESIGN` dict (targets + the two free knobs, g_m/I_D and V_S), prints the worksheet of Table 2, and with `--spice` generates a differential-pair deck, runs `ngspice`, and reports predicted-versus-SPICE side by side (Table 3).

Table 2: Sizing worksheet at example operating point ($g_m/I_D = 15$, $V_S = 0.30\,V$, $L = 0.15\,\mu m$, SKY130A `nfet_01v8_lvt`).

Quantity	**Value**	**Units**
R_D (from BW)	212	Ω
g_m required (from gain)	14.9	mS
I_D / I_{tail}	0.99 / 1.99	mA
$V_{ov} = 2/(g_m/I_D)$	0.133	V
current density J_D	14.6	$\mu A/\mu m$
width W (per device)	68.0	μm
output resistance r_o	941	Ω
intrinsic gain $g_m r_o$	14.0	—
transit frequency f_T	51.3	GHz
bias nodes V_D / V_{DS}	0.989 / 0.689	V
power $V_{DD} \cdot I_{\text{tail}}$	2.38	mW

5 The r_o-Limited Gain

Equation (2) assumed $r_o \gg R_D$, but at minimum length this does not hold. The lookup returns $r_o \approx 941\Omega$ at $g_m/I_D = 15$, so the effective load is

$$R_L \;=\; R_D \| r_o \;=\; 212 \| 941 \;\approx\; 173\Omega, \tag{6}$$

and the realised gain and bandwidth become

$$A_{v0} = g_m R_L \approx 2.58 \text{ V/V } (8.23\text{dB}), \qquad f_{-3dB} = \frac{1}{2\pi R_L C_L} \approx 184\text{MHz}. \tag{7}$$

Because R_D is pinned at 212 Ω by the fixed C_L and the bandwidth target (Eq. (1)), and r_o at this length is comparable to R_D, the gain + BW targets are in competing: the finite r_o pulls the gain down to $\sim 8.2dB$ while pushing the bandwidth up past $150MHz$. A single resistively-loaded stage at $L = 0.15\mu m$ cannot simultaneously meet $10dB$ and $150MHz$ for the selected R_D.

Figure 1 makes this concrete: sweeping the inversion level with g_m held at the gain-target value ($14.9\,mS$), the realised gain plateaus near $8.3dB$ and never reaches $10\,dB$. At $L = 0.15\,\mu m$ the intrinsic $g_m r_o$ is simply too small for r_o to dominate R_D, regardless of bias.

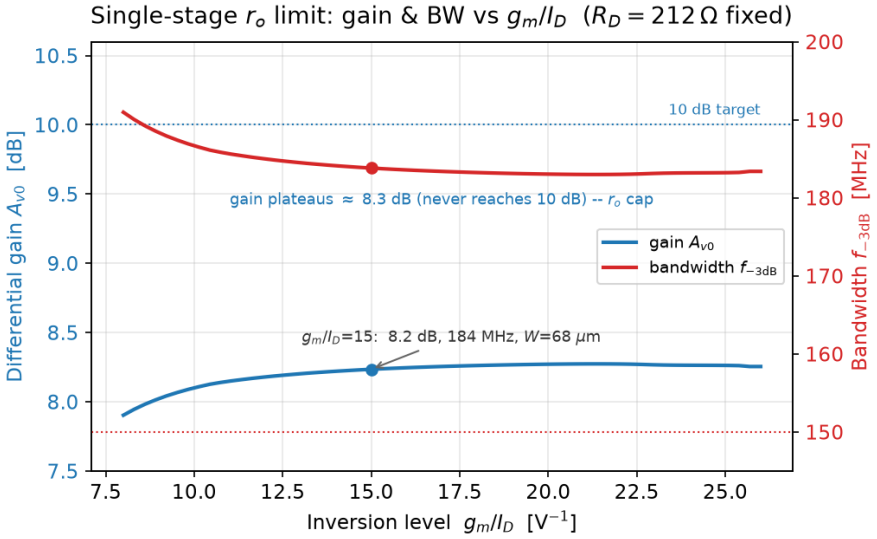


Figure 1: Gain and bandwidth versus inversion level g_m/I_D, with g_m fixed at the $14.9mS$ the $10dB$ target requires and $R_D = 212\Omega$ pinned by the bandwidth. The gain plateaus near $8.3dB$

6 SPICE Verification (SKY130A, tt corner)

The self-contained differential AC testbench `sim/tb_ac.spice` (floating $\pm v_{sd}/2$ drive, 100Ω gate resistors, ideal $2I_D$ tail sink, $5pF$ pad caps) exercises the design under `ngspice` with the SKY130A `tt` models. The hand-calculations track SPICE closely as shown in Table 3. Particularly, the $R_D \| r_o$ gain estimate of Section 5 reproduce the simulated gain to within $0.01dB$, confirming the r_o-limited result is real.

Table 3: Predicted (hand-calc) versus SPICE at the sized operating point ($W = 68\mu m$, $g_m/I_D = 15$).

Quantity	Predicted	SPICE	Units
Tail current I_{tail}	1987	1987	μA
g_m per device	14.9	14.85	mS
Tail-node voltage V_S	0.30	0.205	V
Differential gain A_{v0}	8.23	8.24	dB
Bandwidth f_{-3dB}	183.8	177.8	MHz

The simulated response (Figure 2) is a flat $\sim 8.2dB$ passband rolling off at the single output pole. The responce is both below the $10dB$ gain target and above the $150MHz$ bandwidth target. The tail

node settles at $205mV$ (just above the $200mV$) and the drain DC operating point of $0.988V$ matches $V_{DD} - I_D R_D$ exactly.

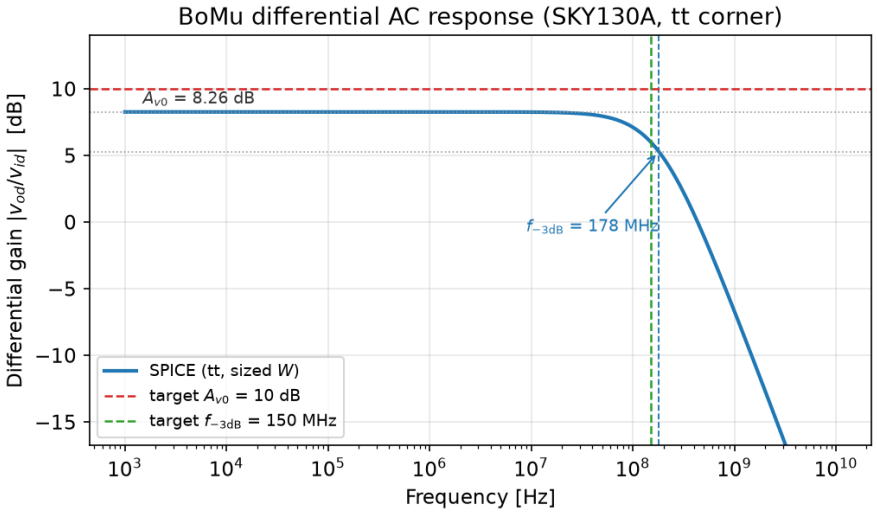


Figure 2: Differential AC response from `sim/tb_ac.spice` (SKY130A, `tt`, sized $W = 68\mu m$).

7 Design-Space Sweeps and the Gain–Bandwidth Limit

Section 5 showed that at the chosen $R_D = 212\Omega$, the finite r_o leaves the gain short and the bandwidth long. This section sweeps R_D and L to map the achievable space.

7.1 R_D: trading gain for bandwidth at constant $g_m/(2\pi C_L)$

For a single output pole the gain–bandwidth product is

$$A_{v0} \cdot f_{-3dB} = g_m(R_D \| r_o) \cdot \frac{1}{2\pi(R_D \| r_o)C_L} = \frac{g_m}{2\pi C_L}, \tag{8}$$

With $g_m = 14.9mS$ and $C_L = 5pF$ this ideal product is $474MHz$. Sweeping R_D (Figure 3) therefore slides the design along this fixed hyperbola. A larger R_D buys gain at the cost of bandwidth. This sweep is at constant power, since R_D draws no bias current.

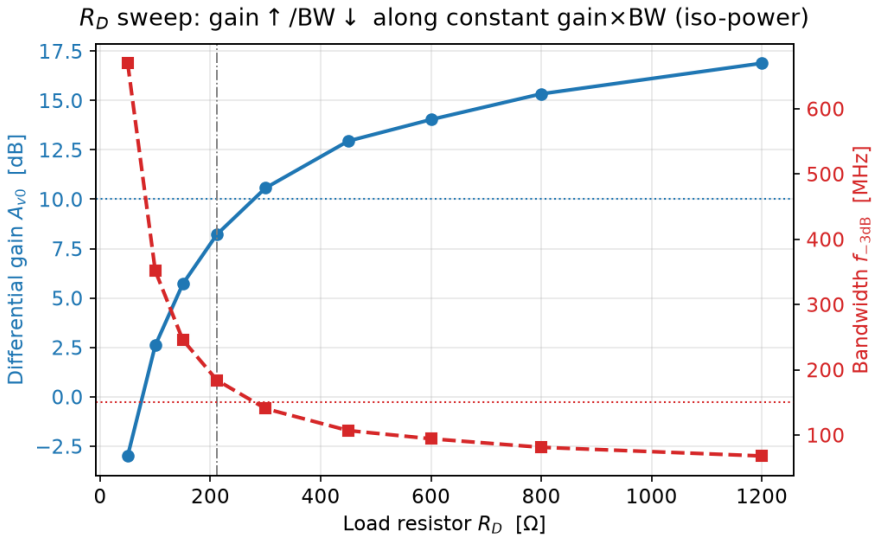


Figure 3: R_D sweep (iso-power $2.38mW$): gain rises and bandwidth falls along the constant $g_m/(2\pi C_L)$ product. Dotted lines are the $10dB$ and $150MHz$ targets.

Channel length was also swept in an attempt to recover additional gain. Longer L raises r_o, so $R_D \| r_o \to R_D$ and the gain climbs toward the ideal $g_m R_D$ at constant power (the current is unchanged, $W \propto L$ to hold the inversion level). The recovery is modest and saturating: $8.2dB$ at $L = 0.15\mu m$ up to only $\approx 9.1dB$ by $L = 0.6\mu m$, which was still short of the $10dB$ target confirming the r_o-limited ceiling of Section 5. Length trades area for a fraction of the missing gain but does not, on its own, reach the target.

7.2 Is $R_D \approx 280\,\Omega$ feasible for $10\,dB$ and $150\,MHz$?

The two targets have product $3.162 \times 150MHz = 474MHz$, which is exactly the ideal GBW of Eq. (8). The specification therefore sits right on the single-stage limit, reachable only if the realised product equals the ideal. It does not: second-order gate/Miller poles pull the SPICE gain–bandwidth product down to $\approx 467MHz$, about 1.5% short. Table 4 brackets R_D in SPICE (`tt`, $L = 0.15$, $W = 68$), note no value of R_D has sufficient GBW product at the chosen g_m.

Table 4: R_D bracket in SPICE.

R_D [Ω]	A_{v0} [dB]	f_{-3dB} [MHz]	GBW [MHz]
220	8.49	175.8	468
250	9.37	158.7	467
265	9.76	**151.7**	467
280	**10.13**	145.4	467
300	10.58	137.9	466

So $R_D \approx 280\Omega$ is a close but still not a feasible solution: the gain passes ($10.13dB$) but the bandwidth is $145MHz$ ($\sim 3\%$ low). Moving to $R_D = 265\Omega$ recovers bandwidth to $152MHz$ but drops the gain to $9.76dB$. The $A_{v0} = 10dB$ crossover lands at $R_D \approx 277\Omega$ with only $f_{-3dB} \approx 148MHz$. The design lands within $\sim$ 2–3% of both but meets neither pair simultaneously, with no margin. Considering this is the nominal `tt`/27°C case (slow corners erode g_m, and V_{tail} sits at the $0.20\,V$ compliance edge in this R_D range), a small margin will be require to meet the design targets after parasitic extraction. Closing the last few percent requires more GBW which can be achieved with a slighty higher g_m (more current/power).

7.3 Meeting both specs: a small g_m increase

Because the realised GBW scales with g_m (Eq. (8)), the $\sim 1.5\%$ shortfall is closed by a comparably small increase in g_m. Holding $g_m/I_D = 15$ and $L = 0.15\mu m$, raising g_m from 14.9 to $16.0mS$ (widening the pair

to $W = 73\mu m$ and the tail to $I_{tail} = 2.13mA$) lifts the realised gain–bandwidth product to $\approx 500MHz$, clearing the $474MHz$ the pair requires. Re-centring the load at $R_D = 262\Omega$ then lands both targets with margin (Table 5): $A_{v0} = 10.15dB$ and $f_{-3dB} = 155.5MHz$ at $2.56mW$ — a 7.6% power increase over the $2.38mW$ baseline. The tail node sits at $0.218V$ (with $V_{SC} = 0.92V$), clear of the $0.20V$ compliance budget.

Table 5: Closing the GBW gap with a g_m increase, SPICE `tt`, $L = 0.15\mu m$, $g_m/I_D = 15$.

g_m [mS]	W [μm]	R_D [Ω]	A_{v0} [dB]	f_{-3dB} [MHz]	P [mW]
14.9 (base)	68	212	8.24	178.0	2.38
15.6	71	272	10.23	150.2	2.50
16.0	**73**	**262**	**10.15**	**155.5**	**2.56**
16.0	73	272	10.40	151.1	2.56

The bolded row above meets both specifications. The margins ($+0.15dB$, $+5.5MHz$) are comfortable at nominal `tt`/27°C but modest. A slow/hot corner erodes g_m, and layout parasitics will erode the bandwidth further. Section 7.4 addresses this directly, sizing a small additional current margin against those parasitics to fix the final operating point.

7.4 Layout-parasitic headroom and the pre-layout operating point

The $g_m = 16mS$ point marginally meets both targets (f_{-3dB} is only $5.5MHz$ over target), and layout will add output-node capacitance (drain-junction C_{db} plus pad interconnect) on top of the fixed $5pF$. Since $f_{-3dB} = 1/(2\pi R_L C_{tot})$ decreases with total capacitance while the DC gain $g_m R_L$ remains unaffected, only the bandwidth spec is exposed. The remedy is a small $\sim 7\%$ g_m margin: widening the pair to $W = 78\mu m$ ($I_{tail} = 2.28mA$) and re-centring $R_D = 245\Omega$ lifts the nominal bandwidth to $166MHz$ at $10.16dB$ / $2.74mW$, holding both specs past $500fF$ of added per-output capacitance.

7.5 Tail-current mirror and input common-mode

The ideal tail sink is realised on-chip as a simple two-transistor resistor-programmed NMOS mirror, kept fully internal. A cascode/Wilson mirror needs $V_{GS} + V_{DS,sat} \gtrsim 0.5V$ of compliance, far above the tail budget, so only a simple mirror fits. The $0.218V$ tail of the sizing above is too tight for a $2.28mA$ sink, but the pair has abundant unused V_{DS} (drains at $0.9V$, sources at the tail): raising the input common-mode from $V_{SC} = 0.92$ to $1.05V$ lifts the tail to $\approx 0.33V$, putting the mirror in clean saturation with a device roughly half the width. The $400mV_{pp}$ swing survives (lowest single-ended output $0.82V$, still $0.32V$ above the saturation floor).

The mirror uses standard-V_T `nfet_01v8` devices at $L = 0.5\mu m$: output $M_3 = W80\mu m/nf8$, reference $M_4 = W10\mu m/nf1$, ratio $N = 8$, so the reference leg draws only $I_{tail}/8 \approx 0.285mA$ through $R_{ref} \approx 745\Omega$ from V_{DD}. An integrated simulation (`sim/bias.py verify`) with the real mirror replacing the ideal sink closes the operating point and leaves the differential spec untouched (Table 6): at $R_{ref} = 745\Omega$ the tail settles at $0.331V$, $I_{tail} = 2.28mA$, $A_{v0} = 10.13dB$, $f_{-3dB} = 167MHz$. The gain only degraded by $0.035dB$ from the ideal source, as the mirror's finite r_{out} lands on the common-mode path (CMRR $\approx 30dB$), not the differential one. Power is $3.08mW$. Because the reference is a plain resistor from V_{DD}, I_{tail} tracks supply/temperature/process — the axis that dominates the PVT sweep of Section 8. The mirror is folded into `ip/bomu.sch` (M_3/M_4, R_{ref}).

Table 6: Integrated verification (`sim/bias.py verify`): BoMu with the real tail mirror ($W_{out} = 80$, $W_{ref} = 10$, $L = 0.5$, $N = 8$), sweeping the programming resistor. $V_{SC} = 1.052V$.

R_{ref} [Ω]	V_{tail} [V]	I_{tail} [mA]	A_{v0} [dB]	f_{-3dB} [MHz]
720	0.330	2.322	10.19	167.5
745	0.331	2.284	10.13	167.1
760	0.332	2.261	10.09	166.8

8 PVT Corner Sign-off and the Final Re-size

The operating point of Sections 7.4–7.5 ($g_m = 17.1mS$, $W = 78\mu m$, $R_D = 245\Omega$) meets both specifications, but only at the nominal `tt`/27°C/1.20V point. A proper sign-off requires the full process–voltage–temperature (PVT) check. The pair is swept over a 63-point grid — seven process corners $\{\texttt{tt}, \texttt{ss}, \texttt{ff}, \texttt{ss_ll}, \texttt{ss_hh}, \texttt{ff_ll}, \texttt{ff_hh}\}$ (the `_ll`/`_hh` suffixes are the poly-resistor process skews) crossed with $T \in \{0, 27, 70\}$°C and $V_{DD} \in \{1.14, 1.20, 1.26\}V$ (±5%) — driven off the extracted schematic netlist by `sim/corners.py`, so the loads are the actual `res_high_po` devices (which track the resistor corner) and the tail is the real mirror (not the idealised $R_D = 245/R_{ref} = 745$ of the testbench).

At $W = 78\mu m$ the sweep fails PVT badly, only $\sim 12/63$ corners pass. The failure is structural, not a trim: the gain–bandwidth wall $A_{v0} \cdot f_{-3dB} = g_m/(2\pi C_L)$ of Eq. (8), which nominal clears by only $\sim 1.5\%$ while the PVT spread is far wider. Three mechanisms bind. *(i)* Supply: the simple mirror sets $I_{tail} \propto (V_{DD} - V_{GS})/R_{ref}$, so the whole 1.14$V$ column droops in g_m. *(ii)* Temperature: hot always erodes mobility and hence g_m. *(iii)* The resistor corner has a more nuanced impact — gain $\propto R_D$ but bandwidth $\propto 1/R_D$, so the `ll` skew kills gain and the `hh` skew kills bandwidth, and no single R_D centres both. An I_{tail} sweep confirms current alone cannot rescue the slow/hot/low-R corner ($g_m \propto \sqrt{I_D}$: +40% current buys only $\sim 1.5dB$).

Two decisions close the sign-off. First, treat V_{DD} as regulated (a positive-tolerance 1.20V rail), which removes the dominant supply axis. The binding limit then becomes the low-R resistor corner, whose $\sim -16\%$ spread must be covered by raw g_m. Second, brute-force g_m by widening the pair to $W = 320\mu m$ ($nf = 33$) and re-centring the loads at $R_D = 193\Omega$ (a wider `res_high_po_5p73` at $L = 2.21\mu m$), with the mirror programming resistor moved to $R_{ref} \approx 380\Omega$ (`res_high_po_2p85`, $L = 1.74\mu m$) for $I_{tail} \approx 2.8mA$. This is the point folded into `ip/bomu.sch` and carried into layout. Table 7 summarizes it.

Table 7: Signed-off sizing (`ip/bomu.sch`).

Device	**Sizing**	**Role**
M_1, M_2 (pair)	`nfet_01v8_lvt` W=320 μm, L=0.15, nf=33	input diff pair
R_{D1}, R_{D2}	`res_high_po_5p73` L=2.21 μm ($\approx 193\,\Omega$)	drain loads
M_3 (mirror out)	`nfet_01v8` W=80 μm, L=0.5, nf=8	tail sink (N=8)
M_4 (mirror ref)	`nfet_01v8` W=10 μm, L=0.5, nf=1	diode reference
R_{ref}	`res_high_po_2p85` L=1.74 μm ($\approx 380\,\Omega$)	I_{tail} program
I_{tail}	$\approx 2.8\,mA$ (regulated $1.20\,V$)	—

At this sizing the full ±5% VDD grid gives 61/63 passing corners: bandwidth now clears $150MHz$ at every corner (worst $161MHz$ @ `ss_hh`/70/1.14), and the only two misses are marginal gain ($9.64dB$ @ `ss_ll`/70/1.14 and $9.86dB$ @ `ff_ll`/70/1.14), both at the simultaneous $-5\%V_{DD}$ + hot + low-R worst case. At the regulated 1.20V rail all 21 process×temperature points pass, the tightest being `ss_ll`/70/1.20 = $10.53dB$. A VDD-margin sweep places the allowable window at $[1.20,\ \gtrsim 1.38]\,V$: the supply may rise $\sim 15\%$ freely but has essentially zero downward margin, so the 1.20V regulated VDD assumption is a positive-only tolerance. The last $\sim 1dB$ of gain at the low-R/hot corner is what enfoces the VDD tolerance, a direct consequence of the single-stage GBW wall rather than any bias defect.

9 Physical Layout

The `bomu` cell is drawn in `magic` on SKY130A. The overriding layout concern for a differential pair is matching of M_1/M_2, so the core is a common-centroid `ABBAABBA` interdigitation: a 2×8 array of a `unit_nch` cell (each $nf = 4$, $W = 10\mu m$, $L = 0.15\mu m$, on its own isolated diffusion island), 16 units total (64 fingers), assigned to the two halves in an alternating centroid pattern so that a linear process gradient across the block cancels differentially.

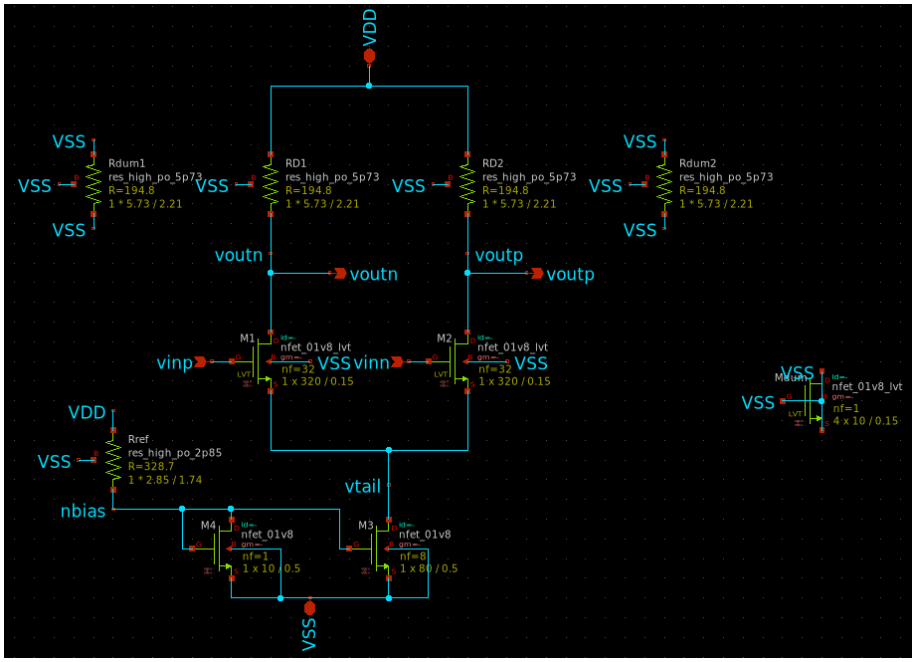


Figure 4: The final `ip/bomu.sch` in `xschem`. This is the schematic the layout implements and LVS-matches.

Edge dummies. Matching is protected at the array edges by same-width $nf = 1$ dummy transistors on the left and right columns (four dummies), which give the outermost live fingers an identical poly/diffusion neighbourhood. A full surrounding ring of ddummy transistors was rejected as it would require$\sim 2.6\times$ the area for second-order benefit, since the isolated-island units already bake in first-order length-of-diffusion matching. The dummy gates, sources and drains are all strapped to `VSS`. The load resistors are handled the same way: the two live loads R_{D1}/R_{D2} (`res_high_po_5p73`) are flanked by two dummy resistors R_{dum1}/R_{dum2} of identical geometry for etch/density symmetry at the resistor-bank edges. These dummies are terminated to `VSS`, giving them a defined path through the p-well rather than leaving them floating.

Tail mirror and bias. The mirror is laid out as a single interdigitated row of nine `nfet_01v8` unit fingers ($W = 10\mu m$, $L = 0.5\mu m$), with the diode-connected reference M_4 at the row centre and the eight M_3 output fingers around it (a 1-D common-centroid, ratio $N = 8$). The programming resistor R_{ref} (`res_high_po_2p85`) sits beside the mirror inside a shared bias guard ring.

Wells and guard rings. Being an all-NMOS block, every sub-block (pair, resistor bank, mirror) is wrapped in a p^+ guard ring tied to `VSS` with no n-well, using a DRC-clean two-frame recipe (a wider `psubdiff` diffusion frame with an inset `psubdiffcont` contact frame under a full `li` cover). The six top-level ports (`vinp`, `vinn`, `voutp`, `voutn`, `VDD`, `VSS`) are brought out on `metal2` (or `metal1` for VSS) past the outer guard ring.

Sign-off. The full block (Figure 5) is clean against `ip/bomu.sch` (Figure 4): `netgen` reports `Circuits match uniquely`, `magic` geometric DRC reports 0 errors, and `klayout sky130A_mr` (`feol` + `beol` + `offgrid` + `floating_met`) reports `CLEAN`. The signed-off GDS and the LVS/DRC reports are collected in `tapeout/` with a checksum manifest (`tapeout/SIGNOFF.md`).

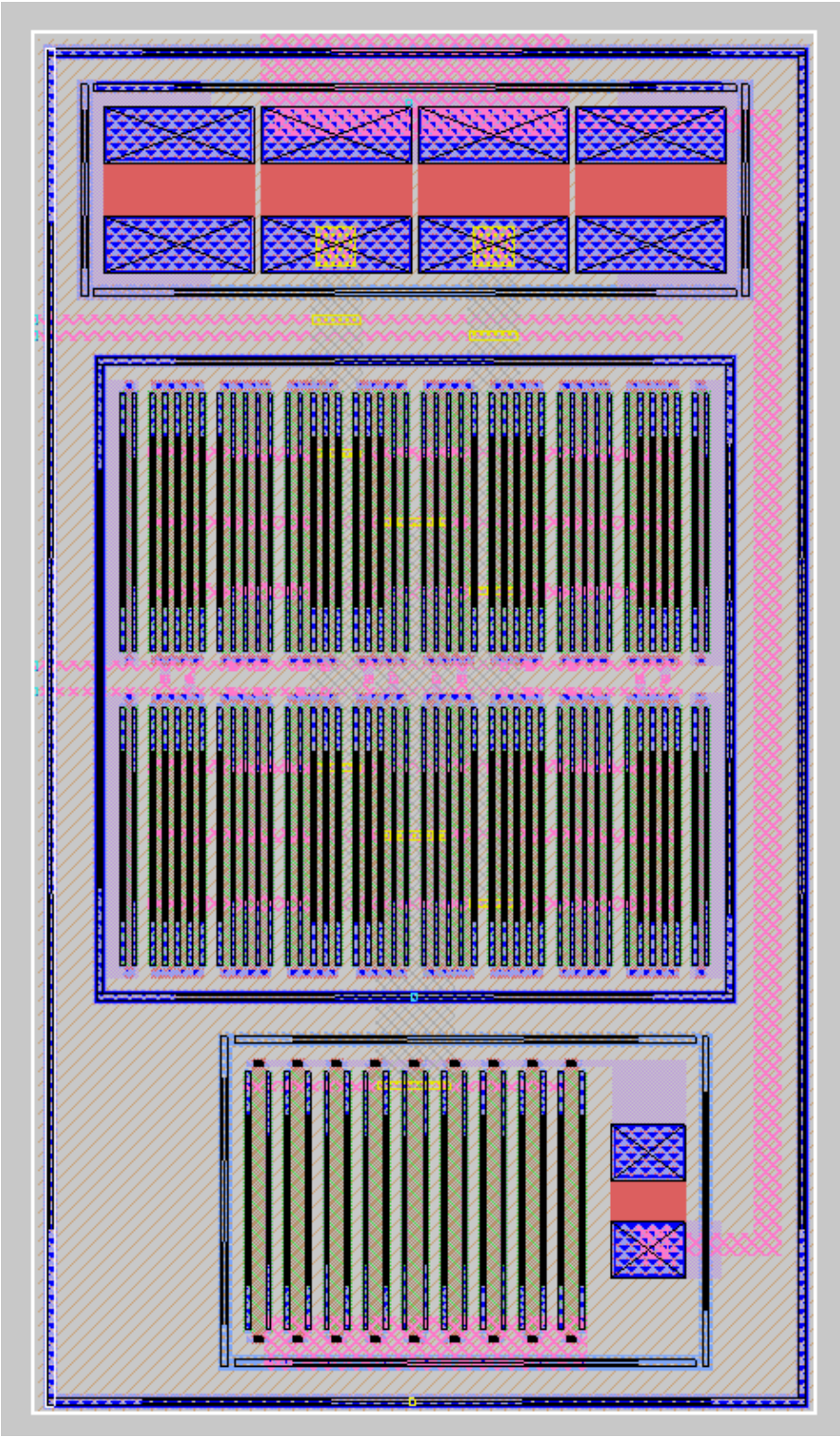


Figure 5: The signed-off `bomu` layout in `magic`: the common-centroid `ABAB` pair core with edge dummies, the flanked load-resistor bank, and the interdigitated tail mirror, each wrapped in a `VSS`-tied p^+ guard ring. DRC- and LVS-clean against the schematic of Figure 4.

10 Post-Layout Extraction (PEX/RCX) and Re-Simulation

With the layout LVS/DRC-clean, the parasitics are extracted from `layout/bomu.mag` and the PVT sweep re-run against the extracted netlist to confirm the sign-off survives layout. Two extractions are

produced (both in `verify/`): a cap-only PEX (`ext_pex.tcl` → `bomu_pex.spice`, 58 coupling/ground capacitors) and a full $R+C$ RCX (`ext_rcx.tcl` → `bomu_rcx.spice`, 75 resistors + 60 capacitors). The RCX flow runs `magic`'s `extresist` on the top-cell interconnect only: recursing into the PDK device cells segfaults on the poly-resistor's internal net, and the device-internal resistance is already in the device models, so only the `bomu` routing needs a wire-resistance network. The extracted subckt port order (`VSS voutn voutp VDD vinp vinn`) differs from the schematic, and the substrate folds into `VSS` with no separate port. The PVT driver `sim/corners.py pex` maps the DUT ports accordingly and `.includes` either netlist, re-running the identical 63-point grid of Section 8.

Parasitic penalty. The extracted parasitics cost bandwidth but not gain, exactly as predicted in Section 7.4: the added output-node capacitance (on top of the fixed $5pF$ pad) pushes the $-3dB$ pole down, while the DC gain $g_m R_L$ carries no capacitance and the wire resistance ($\sim 1.7\Omega$, negligible against the $\sim 193\Omega$ load) barely moves it. The parasitic effects are uniform across corners (Table 8): $\Delta A_{v0} \approx -0.01dB$ and $\Delta f_{-3dB} \approx -3.7\%$. Because it is corner-independent, the schematic worst corners map cleanly onto the post-layout result.

Table 8: Schematic (Section 8) versus post-layout RCX sweeps. The parasitic penalty is a uniform $\sim -3.7\%$ bandwidth and $\sim 0\,dB$ gain.

corner/T/V_{DD}	A_{v0}^{sch} [dB]	A_{v0}^{rcx} [dB]	f_{-3dB}^{sch} [MHz]	f_{-3dB}^{rcx} [MHz]	Δf
`tt`/0/1.14	12.21	12.20	218.9	210.9	-3.7%
`tt`/27/1.20	12.50	12.48	220.2	212.2	-3.6%
`tt`/70/1.14	10.91	10.89	205.5	198.0	-3.7%
`ss`/70/1.14	10.59	10.58	185.8	178.8	-3.8%
`ff`/0/1.20	12.26	12.25	275.4	265.3	-3.7%

Post-layout PVT result. The full RCX sweep gives 61/63 passing PVT corners. This is the same verdict as the schematic: bandwidth clears $150MHz$ everywhere (worst $155.1MHz$ @ `ss_hh`/70/1.14, its $\sim 3.7\%$ shave still above target), and the only two misses are the identical marginal-gain corners ($9.63dB$ @ `ss_ll`/70/1.14, $9.84dB$ @ `ff_ll`/70/1.14). Table 9 lists the worst-case hottest corners at both the -5% edge and the regulated rail. At the regulated $1.20V$ rail all 21 process×temperature points pass post-layout (tightest `ss_ll`/70/1.20 = $10.51dB$, worst BW `ss_hh`/70/1.20 = $158.3MHz$). Peak current acrooss the corners is $\sim 5.2mA$ at `ff_ll`/1.26. The parasitics therefore do not move the sign-off verdict at the operating point — the design is confirmed against the extracted layout.

Table 9: Post-layout (RCX) at the hottest (70°C) corners. At the regulated $1.20V$ rail every corner passes.

	$V_{DD} = 1.14\,V$ (-5%)		$V_{DD} = 1.20\,V$ (reg.)	
corner (70°C)	A_{v0} [dB]	f_{-3dB} [MHz]	A_{v0} [dB]	f_{-3dB} [MHz]
`tt`	10.89	198.0	11.66	203.2
`ss`	10.58	178.8	11.45	182.0
`ff`	10.51	233.6	11.14	242.4
`ss_ll`	**9.63***	212.3	10.51	215.4
`ss_hh`	11.32	**155.1**	12.18	158.3
`ff_ll`	**9.84***	270.6	10.51	279.7
`ff_hh`	11.00	207.0	11.61	215.5

* gain miss at $-5\%V_{DD}$+hot+low-R; passes at the regulated rail.

11 Large-Signal Swing and Distortion

The $400mV_{pp}$ differential output swing (Table 1) is a large-signal requirement, so it is checked with a dedicated transient testbench (`ip/tb_bomu_swing.sch`, Figure 6, which instantiates the `bomu.sym` symbol hierarchically). A differential $1MHz$ sine rides on the V_{SC} input common mode with per-source amplitude $V_a \approx 23.7mV$, so $v_{id,pp} = 4V_a \approx 95mV_{pp}$ drives the amplifier to the swing target. At the `tt` corner ($1.2V$, 27°C) the measured swing is $v_{od,pp} = 391mV_{pp}$, on target, with each single-ended output

spanning 0.79–0.99V about a $\sim 0.8V$ common mode — inside the rails with every device still saturated, so the stage is not clipping.

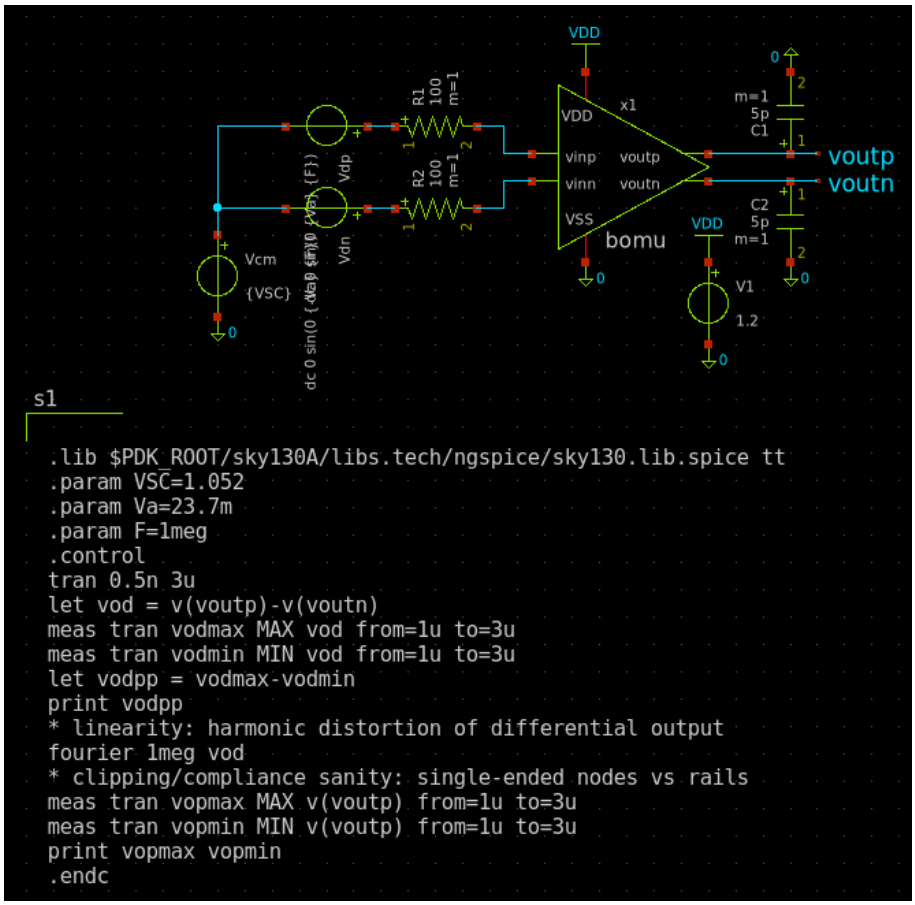


Figure 6: The transient swing testbench `ip/tb_bomu_swing.sch`.

Third-harmonic dominance. A Fourier decomposition of v_{od} (fundamental $1MHz$, ten harmonics) gives THD = 0.53%, set almost entirely by the third harmonic (Table 10). Two features are characteristic of a balanced source-coupled pair. First, the even harmonics vanish (at the $\sim 10^{-14}$ numerical floor): even-order distortion is common-mode and cancels in $v_{od} = v(\text{voutp}) - v(\text{voutn})$, so their absence confirms the devices and loads are symmetric. Second, the residual distortion is odd-order, led by the third: the large-signal transfer $i_{od} \propto \tanh\left(v_{id}/2nV_T\right)$[1] is an odd, compressive function whose Taylor expansion carries only odd powers, so the cubic term folds back as a third harmonic. The fifth and higher harmonics fall off by orders of magnitude, hence THD $\approx$ the third-harmonic term alone. Because that term scales with the cube of the drive, distortion climbs quickly with swing; at the $400mV_{pp}$ operating point it stays a clean 0.53%.

Table 10: Fourier spectrum of v_{od} at the $391mV_{pp}$ operating point (`tt`, $1.2V$, 27°C, $1MHz$ drive).

Harmonic	Freq [MHz]	Norm. mag	
1	1	1.000	fundamental (0.197 V ampl.)
2	2	3×10^{-14}	even (cancelled)
3	3	5.3×10^{-3}	dominant ($\approx$ THD)
4	4	2×10^{-14}	even (cancelled)
5	5	3.6×10^{-6}	
7	7	1.8×10^{-7}	

[1]The tanh is exact for a differential pair whose devices follow the moderate/weak-inversion exponential law $i_D \propto \exp(v_{GS}/nV_T)$. With $V_T = kT/q$ the thermal voltage and n the subthreshold slope factor.

12 Small-Signal Quality: Offset, CMRR, PSRR and Noise

Beyond the headline gain/bandwidth/swing targets, the pair was characterized for input offset, common-mode and supply rejection, and noise, at `tt`/$1.20V$/ 27°C via `sim/corners.py mc` and `sim/corners.py acq`. One caveat frames every mismatch-dependent number: sky130A's open `ngspice` models carry mismatch slopes but never apply them (no `agauss`/`mc_mm_switch` term in the model expressions, and the device subckt ignores `delvto`), so there is no native statistical Monte Carlo. Mismatch is therefore injected with a Pelgrom model using an assumed $A_{VT} = 4mV \cdot \mu m$. The absolute offset/CMRR/PSRR figures scale with that coefficient and should be re-run against a measured A_{VT} for a more accurate calculations.

Input offset. A 200-draw Monte Carlo (per-device V_{th} offset plus 0.5% load-resistor mismatch, run as a single-invocation DC-op loop) gives $\sigma(V_{os}) = 0.88mV$ ($3\sigma = 2.65mV$, worst $2., mV$), centred on zero. The offset is small by construction: the pair is a W=320μm × L=0.15μm = 48μm^2 device, so the Pelgrom term $\sigma_{V_{th}} = A_{VT}/\sqrt{WL} = 0.58mV$ per device ($\sqrt{2}$ for the pair gives $0.82mV$). The large input devices that the sizing already demanded make offset a non-issue.

CMRR and PSRR. Because the output is taken differentially between matched halves, common-mode input and supply ripple couple almost entirely into the output common mode and cancel differentially. The differential CMRR and PSRR are matching-limited, not tail-limited, and are infinite for a perfectly symmetric netlist. Evaluated at one 1σ mismatch draw they are CMRR = $86dB$ and PSRR = $70dB$. The underlying common-mode paths set the ceiling and are finite: the tail-limited common-mode gain is $A_{cc} = -17.8dB$, and because R_D couples V_{DD} straight onto each drain the supply-to-common gain is $A_{sc} = -10.7dB$. This is the same balance that suppressed the even harmonics in Section 11.

Noise. The input-referred noise integrated over $10Hz$–$1GHz$ is $47.9\mu V_{rms}$. Comparing against the $\sim 34mV_{rms}$ input drive that produces the $400mV_{pp}$ output swing, gives an input SNR of order $57dB$.

Table 11: Small-signal quality at `tt`/$1.20V$/27°C. Input offset and the differential CMRR/PSRR are mismatch-limited and scale with the assumed $A_{VT} = 4mV \cdot \mu m$.

Metric	Value	Condition
Input offset $\sigma(V_{os})$	0.88 mV	200-draw MC
Input offset 3σ	2.65 mV	200-draw MC
CMRR (differential)	86 dB	at 1σ mismatch
PSRR (differential)	70 dB	at 1σ mismatch
Common-mode gain A_{cc}	−17.8 dB	tail-limited
Supply→CM gain A_{sc}	−10.7 dB	R_D direct coupling
Input-referred noise	$47.9\mu V_{rms}$	$10Hz$–$1GHz$

13 Status and Next Steps

The central result (Section 7) is that the $10dB$ / $150MHz$ specification sits on the single-stage gain–bandwidth wall $g_m/(2\pi C_L)$, which nominal clears by only $\sim 1.5\%$. A small g_m increase and a $\sim 7\%$ parasitic margin give a nominal-passing $W = 78\mu m$ point, but PVT sign-off (Section 8) fails the full 63-corner box, forcing a re-size to $W = 320\mu m$ / $R_D = 193\Omega$ under a regulated-$1.20V$ rail. At that point the schematic passes 61/63 (21/21 regulated). The block is laid out LVS/DRC-clean (Section 9), and the parasitic-extracted (RCX) netlist re-run across all 63 corners (Section 10) reproduces a uniform $\sim -3.7\%$ bandwidth, $\sim 0dB$ gain penalty with all 21 regulated-rail points passing. The signed-off GDS and reports are in `tapeout/`.

Remaining / optional work:

1. The two marginal PVT misses (9.6–$9.8dB$ at $-5\%V_{DD}$+hot+low-R) are the single-stage GBW wall, not a trim: closing them for a symmetric ±5% supply would need either a relaxed $\sim 9dB$ gain spec, a higher nominal rail ($\sim 1.25V$), or a topology change.

2. `sim/tb_ac.spice` is a single-point AC demonstration deck at the pre-PVT nominal ($W = 78\,\mu m$) point. The only remaining idealisation is that its R_{ref} uses a round value rather than the extracted `res_high_po` resistance, which could be trimmed for an exact match, however, this would be mostly cosmetic as both sims already match closely.

References

[1] P. G. A. Jespers and B. Murmann, *Systematic Design of Analog CMOS Circuits Using Pre-Computed Lookup Tables*. Cambridge University Press, 2017.

[2] B. Razavi, *Design of Analog CMOS Integrated Circuits*, 2nd ed. McGraw-Hill, 2017, ch. 4: Differential Amplifiers.